

MMI 801 · EDUCATIONAL INNOVATION FOR THE MATHEMATICAL SCIENCES

MSc in Mathematical Innovation · Open University of Kenya

Module 5: Playful Learning

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LEARNING OUTCOMES

1. Define playful learning and distinguish it from gamification, game-based learning, and entertainment — and explain its particular importance in mathematics education.
2. Design activities that authentically incorporate play and gamification principles to address specific mathematical learning objectives.
3. Evaluate the effectiveness and limitations of play-based learning approaches using research evidence.
4. Design an original gamified mathematics learning experience, justifying its effectiveness with reference to motivation theory and learning science.

1 Module Overview

Mathematics has an image problem. For millions of students, it is the subject most associated with anxiety, failure, and exclusion. Yet the practices at the heart of mathematical thinking — exploring patterns, making conjectures, testing ideas, surprising yourself with unexpected results — are deeply playful in their nature. The mathematician G.H. Hardy described doing mathematics as “playing with ideas”; the problem is that traditional mathematics instruction systematically removes the play.

This module reclaims the playful dimension of mathematics learning — not as a compromise of rigour, but as its precondition.

EINSTEIN ON PLAY AND MATHEMATICS

“Play is the highest form of research.” — Albert Einstein

This is not a romantic sentiment — it reflects something profound about how mathematical knowledge is created and how it is most deeply learned. When students play with mathematical ideas — without the immediate pressure of evaluation, with freedom to explore wrong paths, with the genuine possibility of surprise — they are doing what mathematicians actually do. The challenge for instructional design is to create that freedom without abandoning structure.

2 Defining Playful Learning

A TAXONOMY OF PLAY IN LEARNING CONTEXTS

Play: Activity chosen freely, intrinsically motivated, process-oriented rather than goal-oriented, imaginative, and active (Whitebread et al., 2017).

Playful learning: Instructional design that incorporates the essential qualities of play — choice, intrinsic motivation, exploration, imagination — within a structured educational context that maintains clear learning goals.

Game-based learning (GBL): Using complete games (digital or physical) as the primary vehicle for learning. The game itself is the learning environment.

Gamification: Applying game design *elements* (points, badges, levels, challenges, narrative) to non-game educational activities. The activity remains educational; the game elements provide motivational scaffolding.

Entertainment: Activity whose primary purpose is enjoyment, with learning as a possible byproduct but not the design goal.

The distinctions matter because each has different implications for instructional design, assessment, and the relationship between enjoyment and learning.

3 Why Playful Learning Matters in Mathematics

3.1 The Mathematics Anxiety Problem

Mathematics anxiety — the experience of tension, apprehension, or fear in response to mathematical tasks — affects approximately 20–25% of students globally (Hembree, 1990) and is significantly more prevalent in African educational contexts where examination pressure is intense. Mathematics anxiety:

- Reduces working memory capacity available for mathematical processing.
- Creates avoidance behaviour that compounds over time.
- Is transmitted from anxious teachers to students.
- Is disproportionately experienced by students from marginalised groups.

Playful learning environments reduce anxiety by removing the immediate threat of evaluation, allowing experimentation, and associating mathematical activity with positive emotional experience.

3.2 Motivation Theory: Why Play Works

SELF-DETERMINATION THEORY (Ryan & Deci, 2000) and Play

Self-Determination Theory (SDT) identifies three basic psychological needs whose satisfaction drives intrinsic motivation:

1. **Autonomy:** The experience of choice and self-direction. Playful learning provides choice in how problems are approached.
2. **Competence:** The experience of effectiveness and mastery. Well-designed mathematical games provide appropriate challenge and visible progress.
3. **Relatedness:** The experience of connection to others. Collaborative mathematical games build community.

When all three needs are satisfied, students engage with intrinsic motivation — working on mathematics because it is inherently interesting, not because of external rewards. Excessive use of extrinsic rewards (points, leaderboards, prizes) can *undermine* intrinsic motivation through the overjustification effect — a critical caution for gamification design.

4 Designing Playful Mathematical Experiences

4.1 Principles for Playful Learning Design

Low stakes entry	All students can begin; initial failure is safe	Open problems with multiple entry points
Exploration before explanation	Students investigate before being told	GeoGebra discovery before algebraic proof
Surprise and wonder	Design for “aha” moments	Problems with non-obvious or counterintuitive solutions
Appropriate challenge	Vygotsky’s ZPD; not too easy, not too hard	Adaptive difficulty; student choice of challenge level
Visible progress	Students see their growth	Skill trees; portfolio; mastery milestones
Collaborative possibility	Peer interaction enhances engagement	Partner or group game structures

EXAMPLE: A Gamified Probability Unit — “The Casino of Mathematics”

Context: Form 4 probability unit (Kenya CBC).

Design: Students operate as “mathematicians” for a fictional casino. Their job is to *design fair games* (expected value = 0) and *expose unfair ones*. Over 4 lessons:

- **Lesson 1 (Play):** Students play 10 simple games using dice, cards, and coins. After each game, they record outcomes and begin to suspect which games favour the “house.”
- **Lesson 2 (Analyse):** Students calculate expected values for each game. The “aha”: mathematical confirmation of their intuitions.
- **Lesson 3 (Design):** Groups design their own game, targeting $E[X] = 0$. Peer review: other groups check the mathematics.
- **Lesson 4 (Tournament):** Class casino day; each group presents their game; the class votes on fairness based on mathematical evidence.

Gamification elements: Group points for accurate expected value calculations; badge for the best-designed fair game; narrative embedding (each group is a “casino design firm”).

Learning achieved: Expected value, probability distributions, fairness, bias — all through genuine mathematical investigation embedded in a playful, competitive frame.

5 Digital Mathematical Games

Prodigy	Primary/junior secondary algebra	Adaptive; highly engaging; free for students; caution: rewards may dominate over content
DragonBox	Algebra fundamentals; equation solving	Strong evidence for pre-algebra concept development (Sande et al., 2021)
Desmos activities	Secondary through university	Teacher-designed; game-like structure; rich mathematical tasks
Number Munchers	Number theory; primes; factors	Classic; low-tech; effective for procedural engagement
GeoGebra 3D Games	3D geometry; spatial reasoning	Emerging; high engagement; requires devices

6 Key Concepts and Terminology

Playful Learning	Instruction incorporating play qualities — choice, exploration, intrinsic motivation — within structured educational goals.
Mathematics Anxiety	Negative emotional response to mathematical tasks; reduces working memory and performance.
Self-Determination Theory	Ryan & Deci's theory: intrinsic motivation driven by autonomy, competence, and relatedness.
Overjustification Effect	External rewards can undermine pre-existing intrinsic motivation.
Expected Value	$E[X] = \sum x_i P(x_i)$; the probability-weighted average outcome of a random variable.
Game-based Learning	Using complete games as the primary learning vehicle.
Flow State	Csikszentmihalyi's concept: optimal experience of deep, effortless engagement at the intersection of skill and challenge.

7 Module Assignment

MODULE 5 ASSIGNMENT

Design a new **gamified mathematics learning experience** for a specific mathematical concept, level, and learner context of your choice.

Your design document (max 800 words + any diagrams) must include:

1. **Context:** Level, topic, number of students, available resources.
2. **Learning objective:** One specific, Bloom's-level outcome.
3. **Game design:** Complete description of the game/activity, including rules, materials, roles, and progression.
4. **Gamification elements:** Which mechanics are used (points, badges, narrative, challenge, etc.) and why each was chosen.
5. **Theoretical justification:** Connect your design to Self-Determination Theory and at least one other learning theory from Module 4.
6. **Overjustification risk:** Identify any extrinsic reward elements in your design and explain how you will mitigate the overjustification effect.
7. **Assessment:** How will you assess mathematical learning, not just engagement?

8 Revision Questions

1. Distinguish between playful learning, game-based learning, and gamification. For a lesson on trigonometric ratios at Form 3, describe a different design for each.
2. Explain the overjustification effect. Design a gamified mathematics activity that avoids it, and explain your specific design choices.
3. How does Self-Determination Theory explain why leaderboards sometimes demotivate students? Under what design conditions might a leaderboard *not* undermine motivation?
4. Calculate the expected value of a game where: roll a die; if you roll 6, win KSh 50; if you roll 1, lose KSh 20; otherwise, nothing happens. Is this game fair? Design a modification that makes it exactly fair.
5. A school principal argues: “Mathematical games waste time that should be spent on exam preparation.” Construct a research-based response to this argument.
6. Design a physical (no-technology) playful learning activity for university-level group theory. Justify its theoretical foundations and describe the mathematical understanding it develops.
7. How would you adapt the Casino of Mathematics example (Section 4) for a context where students have no access to dice, cards, or digital devices?

MODULE 5 SUMMARY

- Playful learning restores the exploratory, intrinsically motivated quality of genuine mathematical thinking that traditional instruction suppresses.
- Mathematics anxiety affects a significant minority of students and is reduced by low-stakes playful learning environments.
- Self-Determination Theory explains why well-designed play produces intrinsic motivation: it satisfies autonomy, competence, and relatedness needs.
- The overjustification effect warns against overuse of extrinsic rewards in gamification design.
- Effective gamification design embeds mathematical content authentically in the game mechanic — not as decoration but as the core activity.
- *Play is the highest form of research; gamification transforms the struggle of learning into the joy of discovery.*

9 Further Reading

- **Ryan, R.M. & Deci, E.L.** (2000). Self-determination theory and the facilitation of intrinsic motivation. *American Psychologist*, 55(1), 68–78.
- **Hembree, R.** (1990). The nature, effects, and relief of mathematics anxiety. *Journal for Research in Mathematics Education*, 21(1), 33–46.
- **Csikszentmihalyi, M.** (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.